

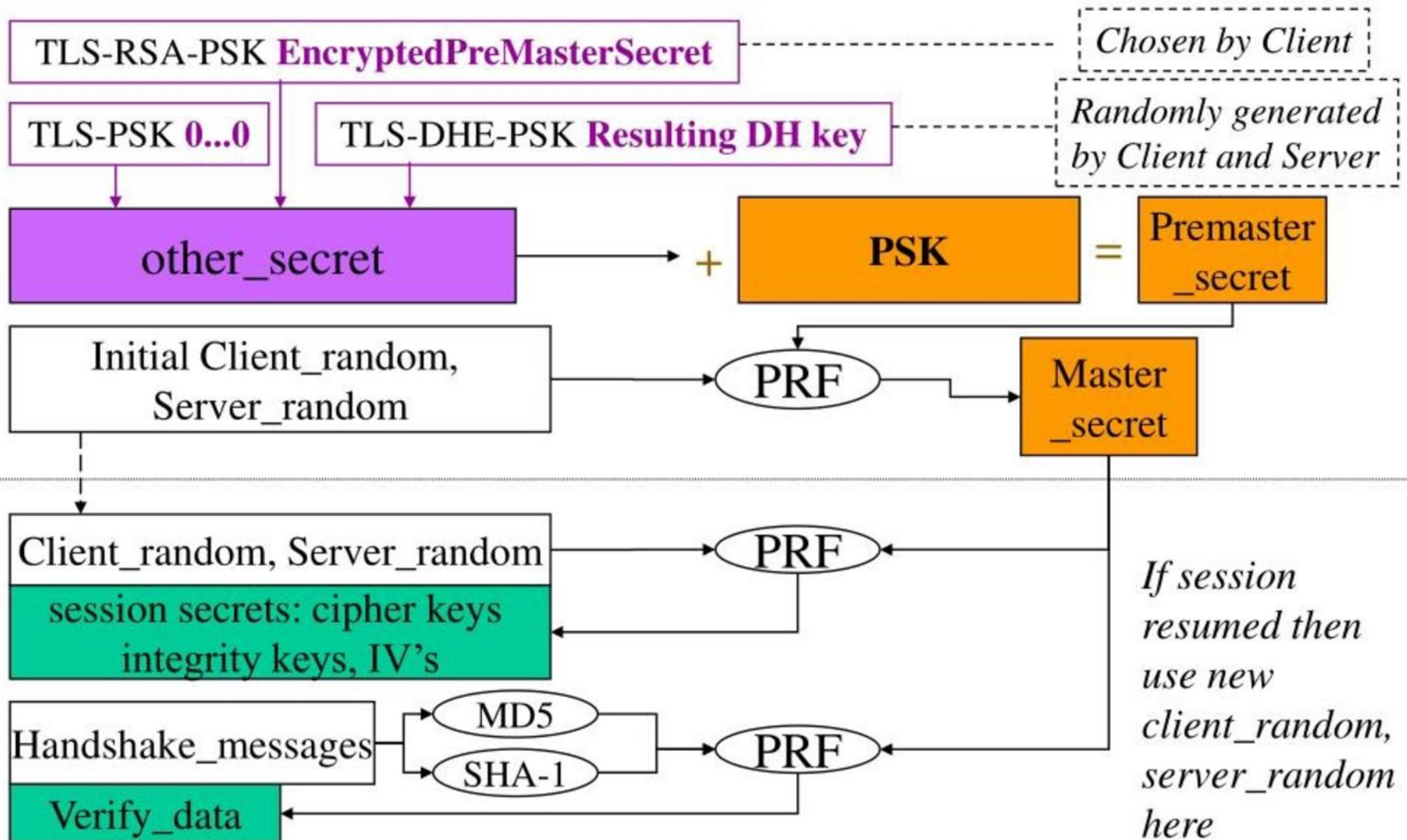
TLS-PSK Handshakes and the UIM

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TLS-PSK Key Generation



Session Secret Generation Model

Computed or Stored in ME

TLS-RSA-PSK **EncryptedPreMasterSecret**

Chosen by ME

TLS-PSK **0...0**

TLS-DHE-PSK **Resulting DH key**

*Randomly generated
by ME and Server*

other_secret

Initial Client_random,
Server_random

Client_random, Server_random
session secrets

Handshake_messages

Verify_data

PSK

Premaster_secret

PRF

Master_secret

PRF

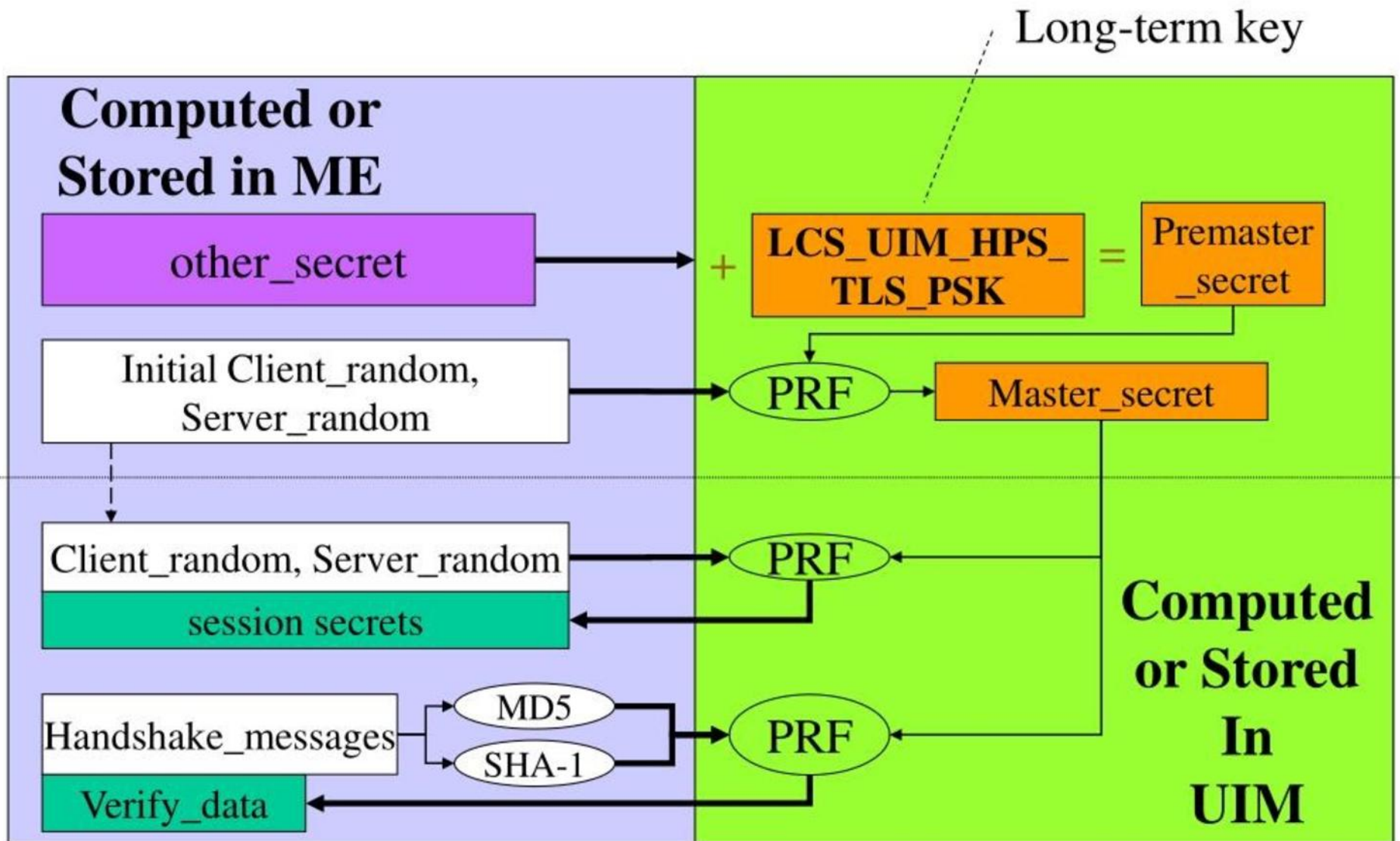
PRF

**Computed
or Stored
In
UIM**

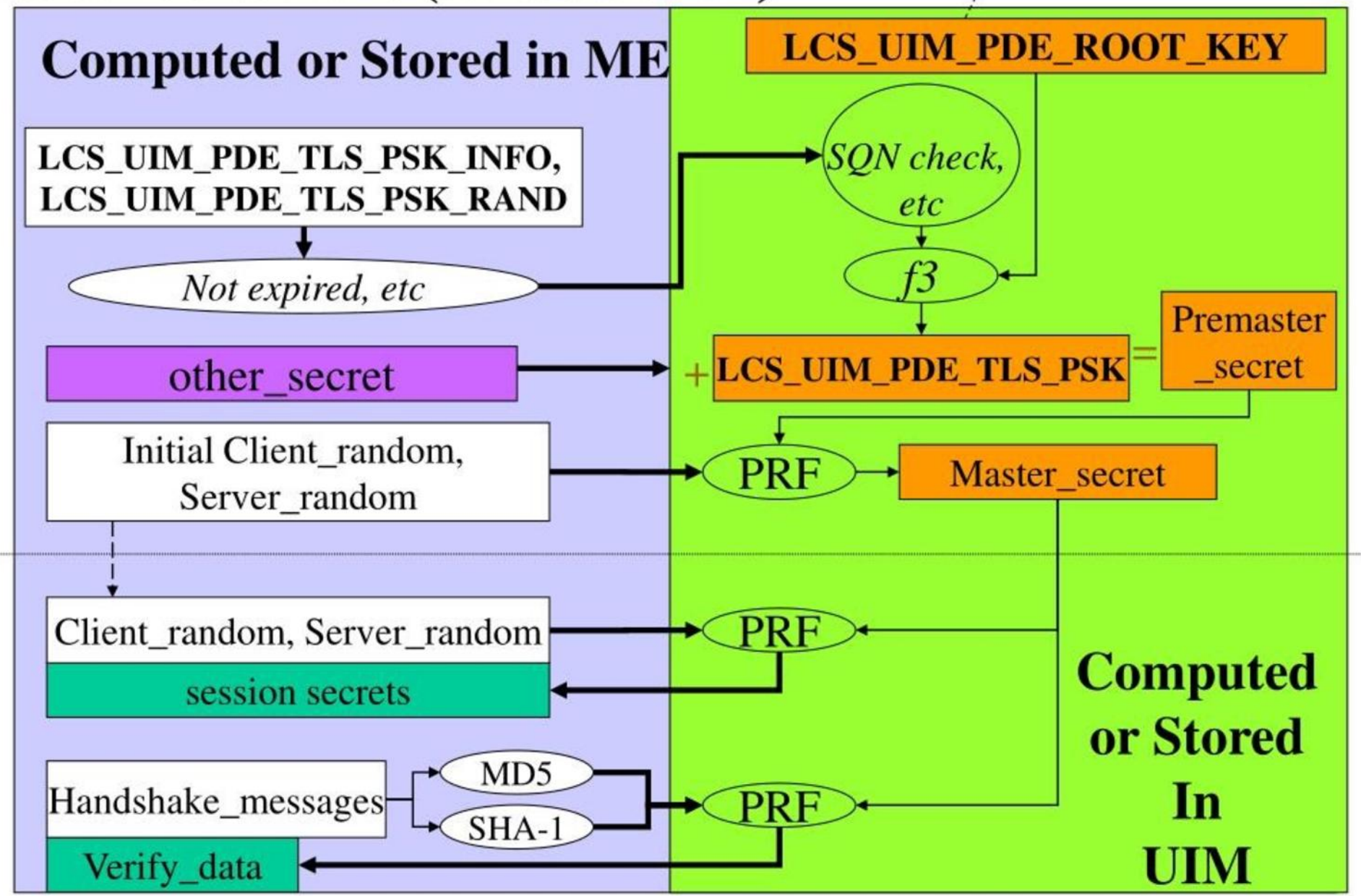
Notes

- UIM doesn't care if pure PSK, or RSA or DHE etc is added.
- UIM must be present for
 - Session secret generation and
 - Verify_data generation
 - Resuming a TLS session
- PSK may be re-used for multiple TLS sessions

MS $\leftrightarrow$ H-PS (Session-A)



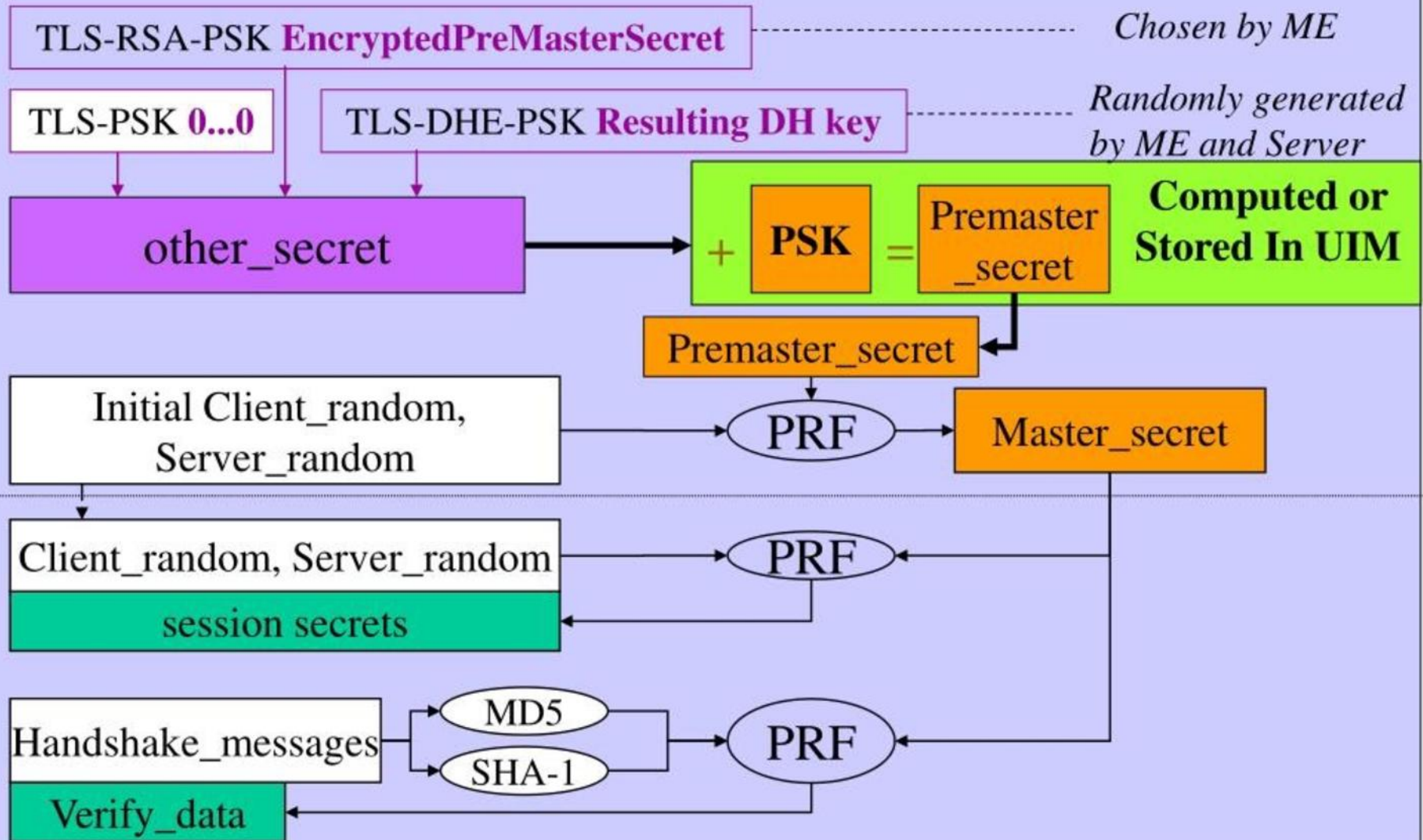
MS ↔ PDE (Session-B)



Why session secrets are
generated in UIM

Model 1 (Very BAD)

Computed or Stored in ME



Problems with Model 1

ME Handshake abilities:

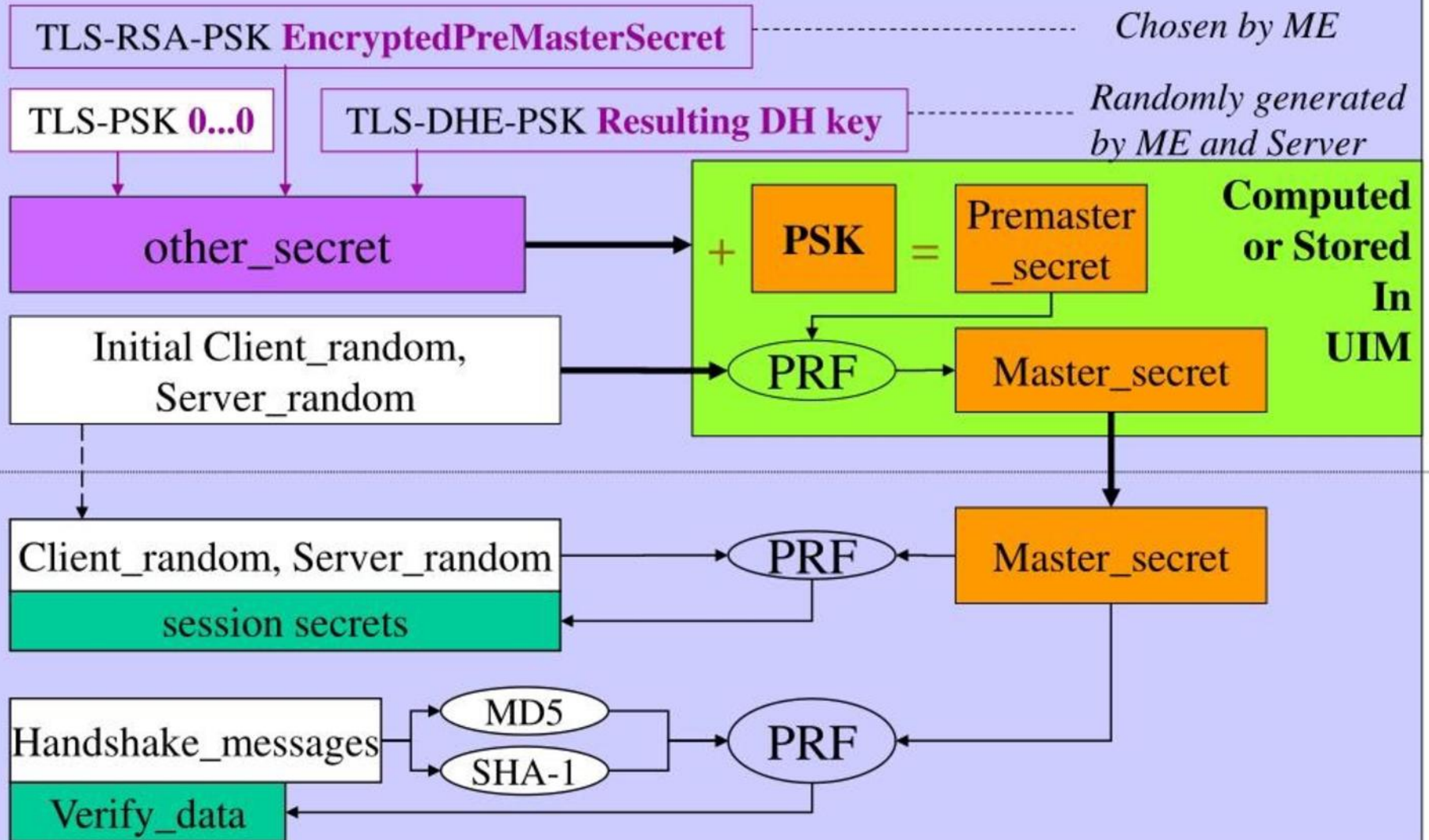
- ME can steal and distribute
 - Master_secret (only used in single TLS session)
 - Premaster_secret (can use in multiple TLS sessions)

Consequences if PSK used more than once:

- Other users can start new TLS sessions: **YES**
- Other users can resume this TLS session: **YES**

Model 2 (BAD)

Computed or Stored in ME



Model 2 Analysis

ME Handshake abilities:

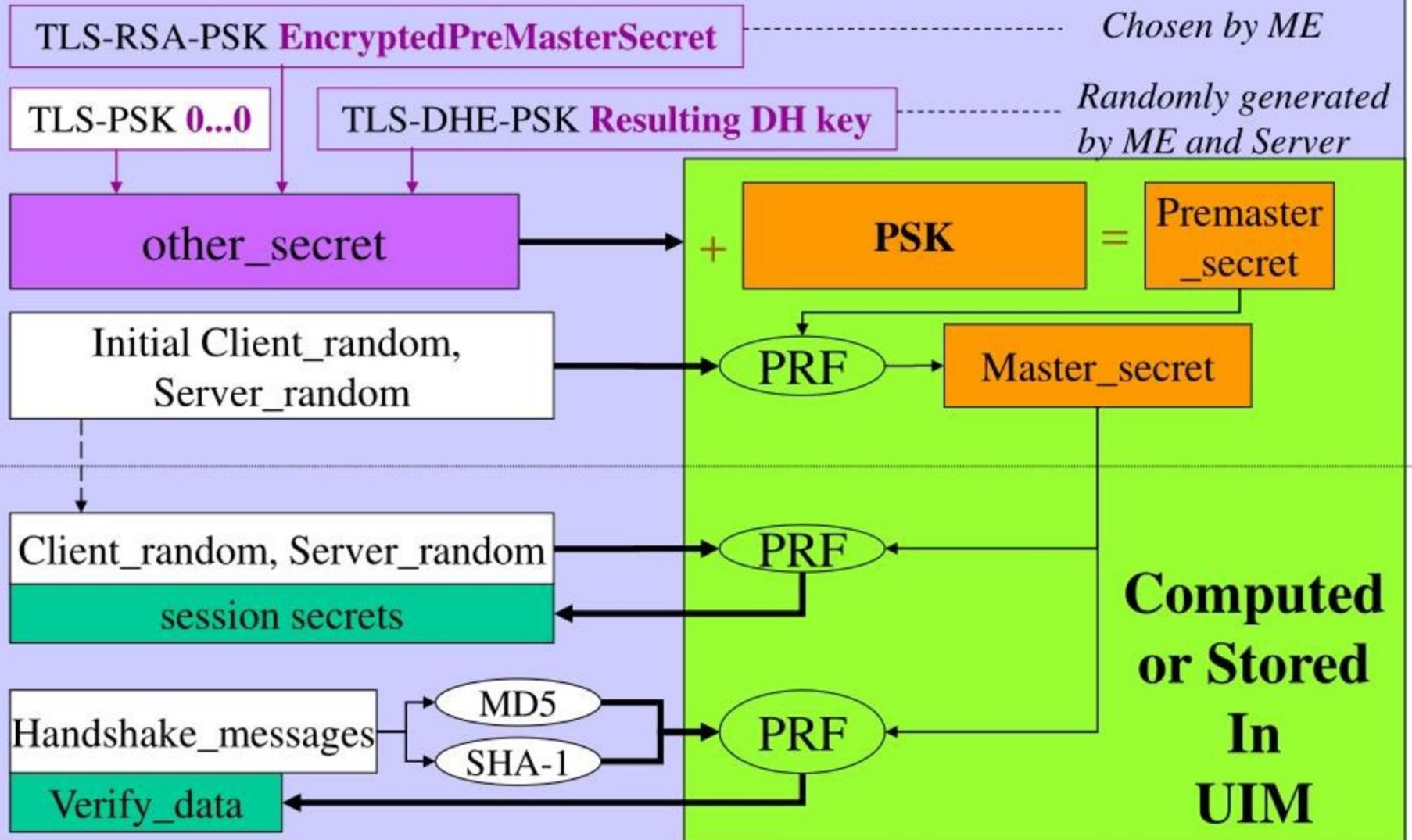
- ME can steal and distribute
 - Master_secret (only used in single TLS session)

Consequences if PSK used more than once:

- Other users can start new TLS sessions: **NO**
- Other users can resume this TLS session: **YES**

Model 3 (GOOD)

Computed or Stored in ME



Model 3 Analysis

ME Handshake abilities:

- None

Consequences if PSK used more than once:

- Other users can start new TLS sessions: **NO**
- Other users can resume this TLS session: **NO**